

July-November 2024 Semester

CS5691: Pattern recognition and Machine Learning Programming Assignment 1

TEAM -25

Team Members:

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- 2. CS24M044 – Sayan Das**

1. Introduction

This report presents the results of the first programming assignment for the course "Pattern Recognition and Machine Learning." The task involved the implementation of linear regression models using polynomial basis functions across different datasets. The models were regularized using quadratic regularization, and the performance was assessed on both training and test datasets.

2. Datasets

Three different types of datasets were used in this assignment:

- 1. Dataset 1:**
 - Type: Univariate (1-dimensional input data)
 - Training Sets: 10 examples (Dataset 1a), 50 examples (Dataset 1b)
- 2. Dataset 2:**
 - Type: Bivariate (2-dimensional input data)
 - Training Sets: 25 examples (Dataset 2a), 100 examples (Dataset 2b)
- 3. Dataset 3:**
 - Type: Multivariate

3. Regression Model and Regularization

- Model: Linear regression model using polynomial basis functions
- Regularization: Quadratic regularization

4. Hyperparameters

Dataset	Degree of Polynomial M	Regularization Coefficient λ
Dataset 1	3, 6, 9	0.001, 0.1, 1
Dataset 2	2, 4, 6	0.001, 0.1, 1
Dataset 3	2, 3	0.000001, 0.0001, 0.1

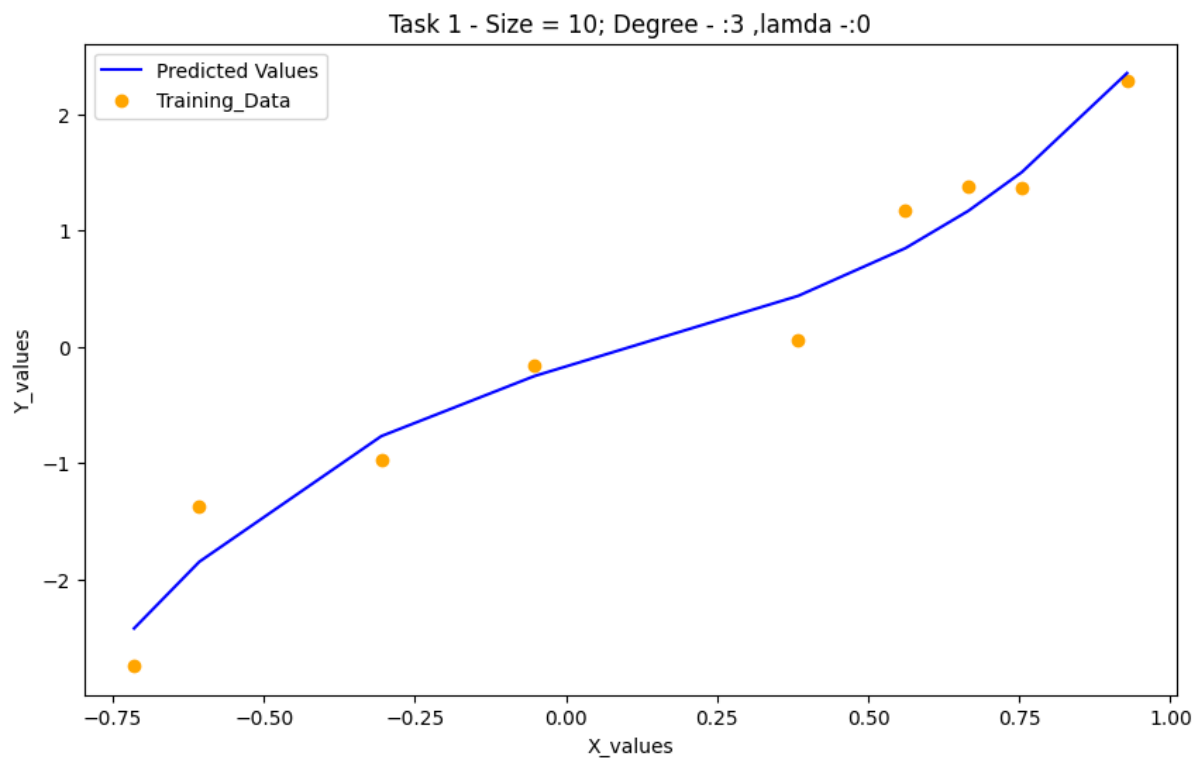
5. Results and Analysis

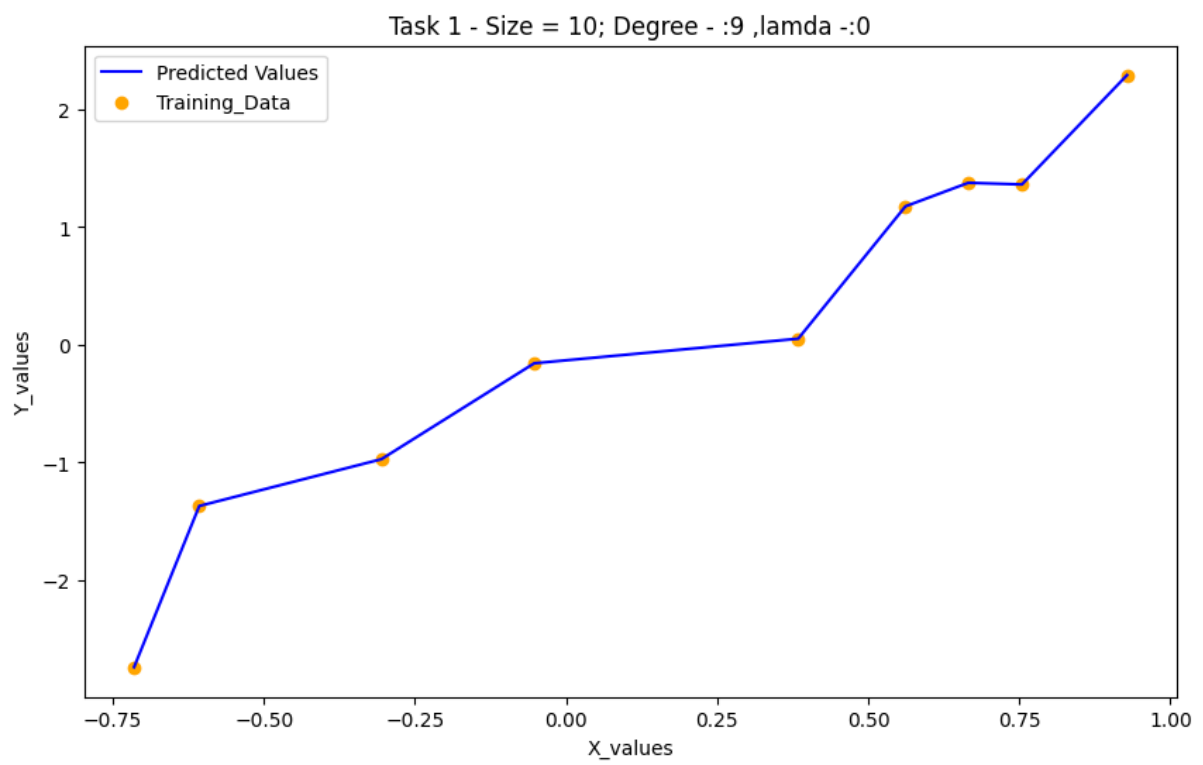
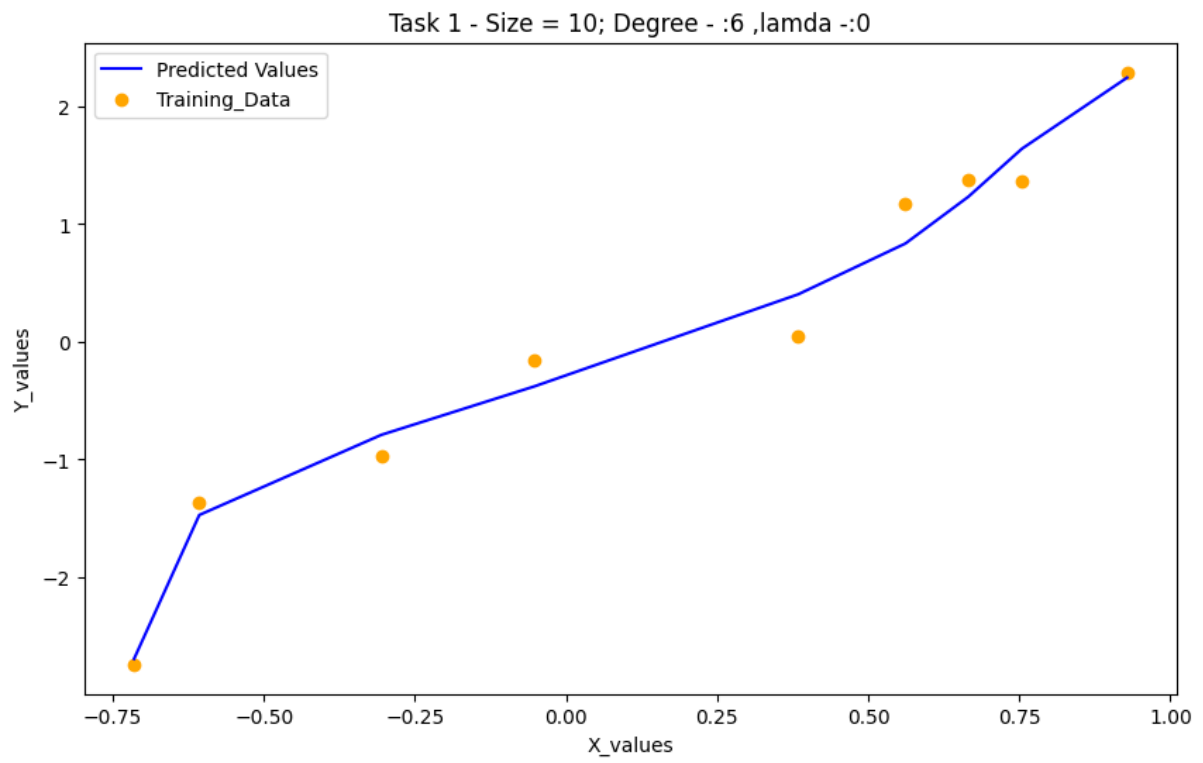
5.1 Dataset 1: Univariate Data

- **Objective1:** To plot the approximated functions (curves) using training datasets of different sizes (10 and 50) for different model complexities with and without regularization, and for different values of λ with model complexity as 9.
- **Plots:** The training data points are superposed on the approximated curves.

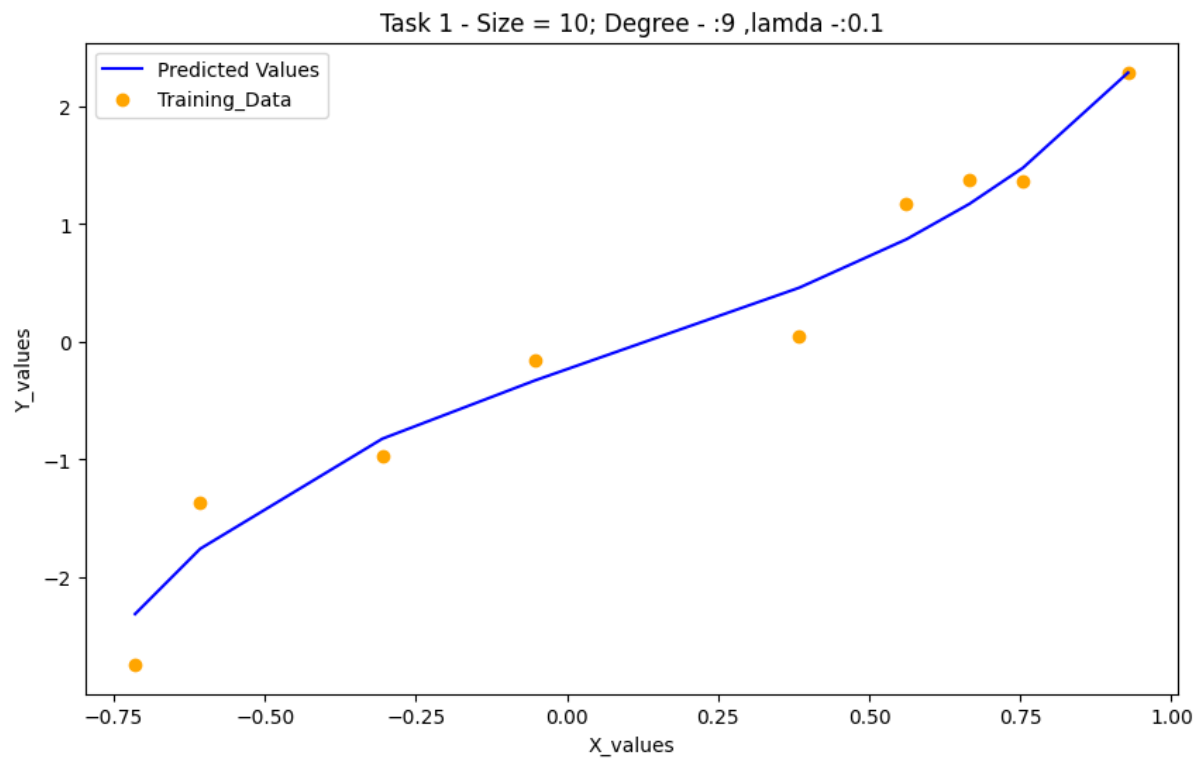
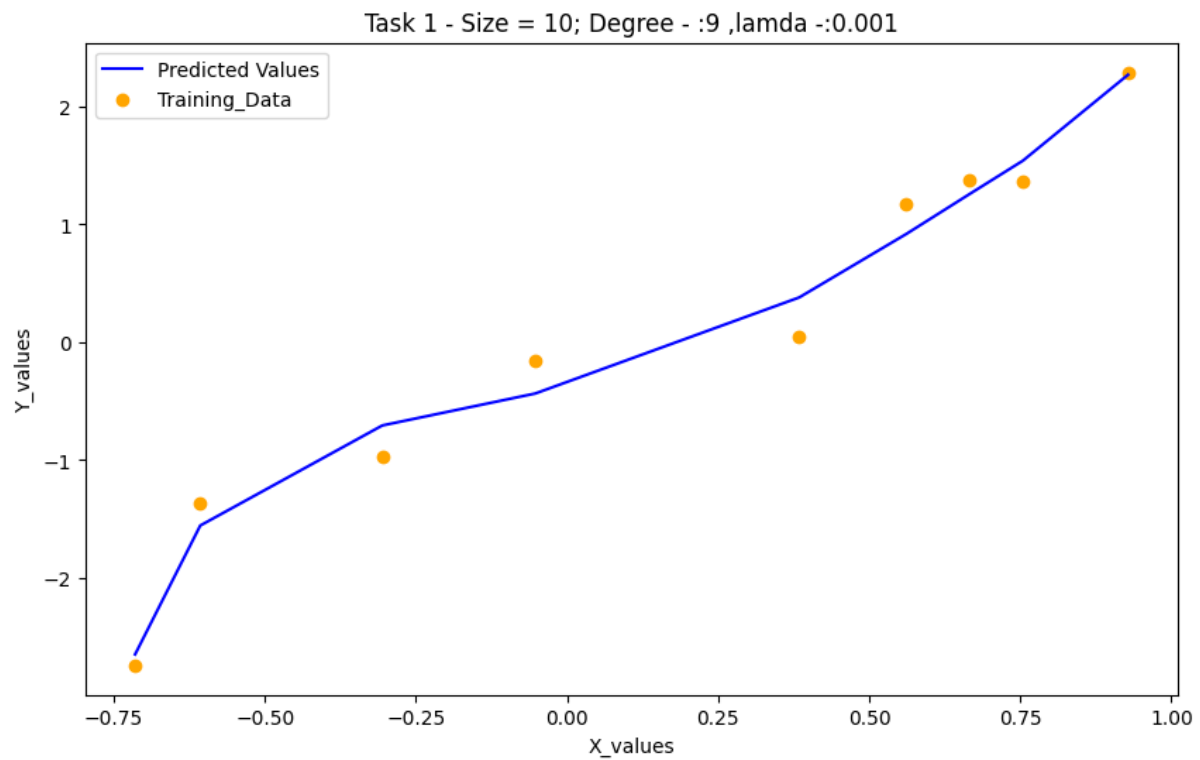
Dataset 1

1. Without Regularization for all degrees :

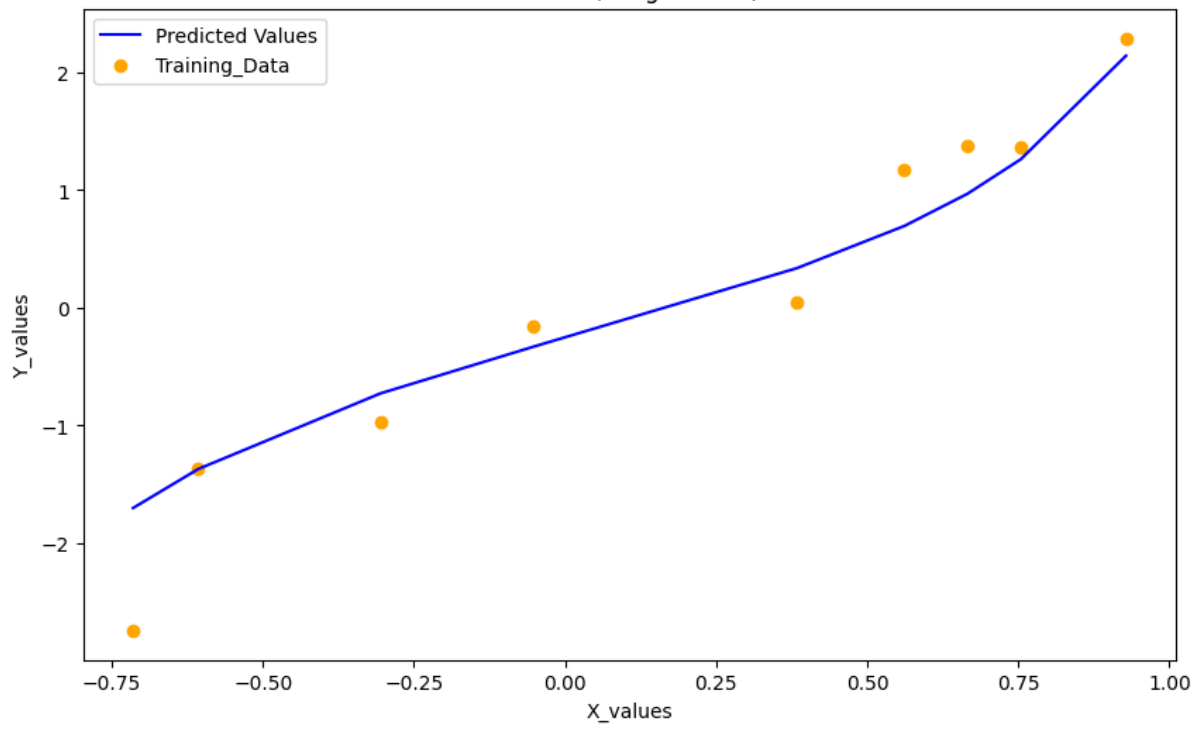




2. With Regularization with degree 9 :



Task 1 - Size = 10; Degree - :9 ,lamda -:1



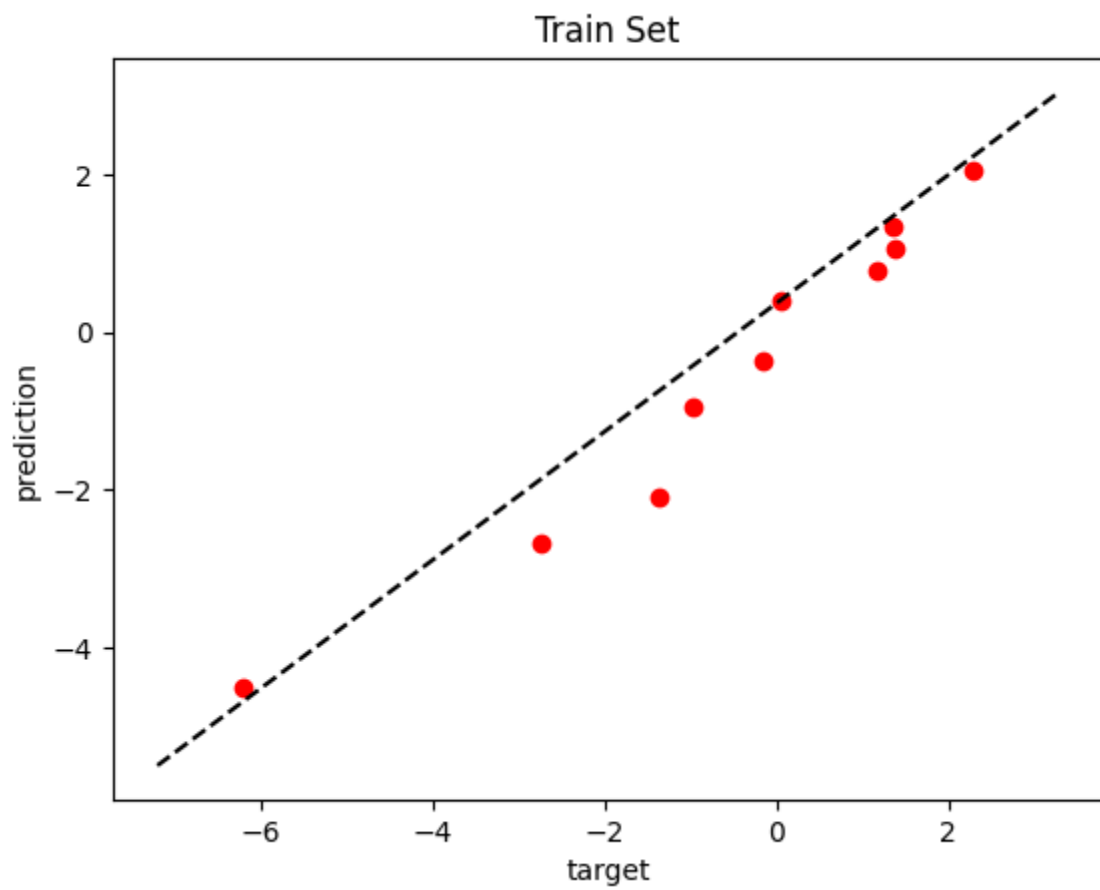
ERMS Table 1(a):

	Without Regulization	Without Regulization	Without Regulization	With Regulization with degree=9	With Regulization with degree=9	With Regulization with degree=9
	M=3	M=6	M=9	Lamda=0.001	Lamda=0.1	Lamda=1
Training	0.2796066761162385	0.21975957282534325	2.620999108221136e-14	0.21368703244376003	0.2784138203964957	0.4316811081551189
Validation	0.47106699595545076	0.6572959953690385	2.1443030172623523	0.5923395547344917	0.4864326647266319	0.48542422279898045
Test	0.4525943078073553	1.1721371936331701	4.900772118303137	0.9784582835558071	0.48054907148497855	0.7465130056268595

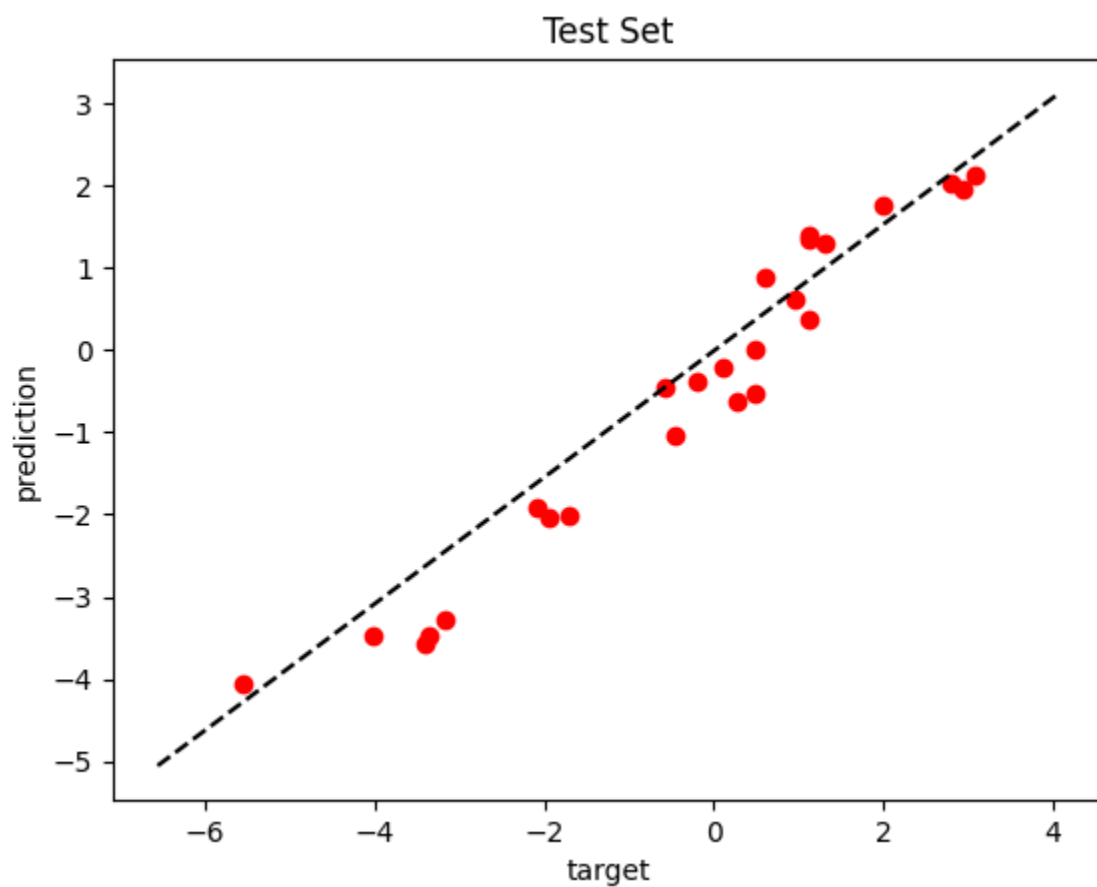
Using above Table best Model with Degree: 3

Lamda: 1 for 1A

Scatter plot for best model



Erms Train: 0.4435629240363046

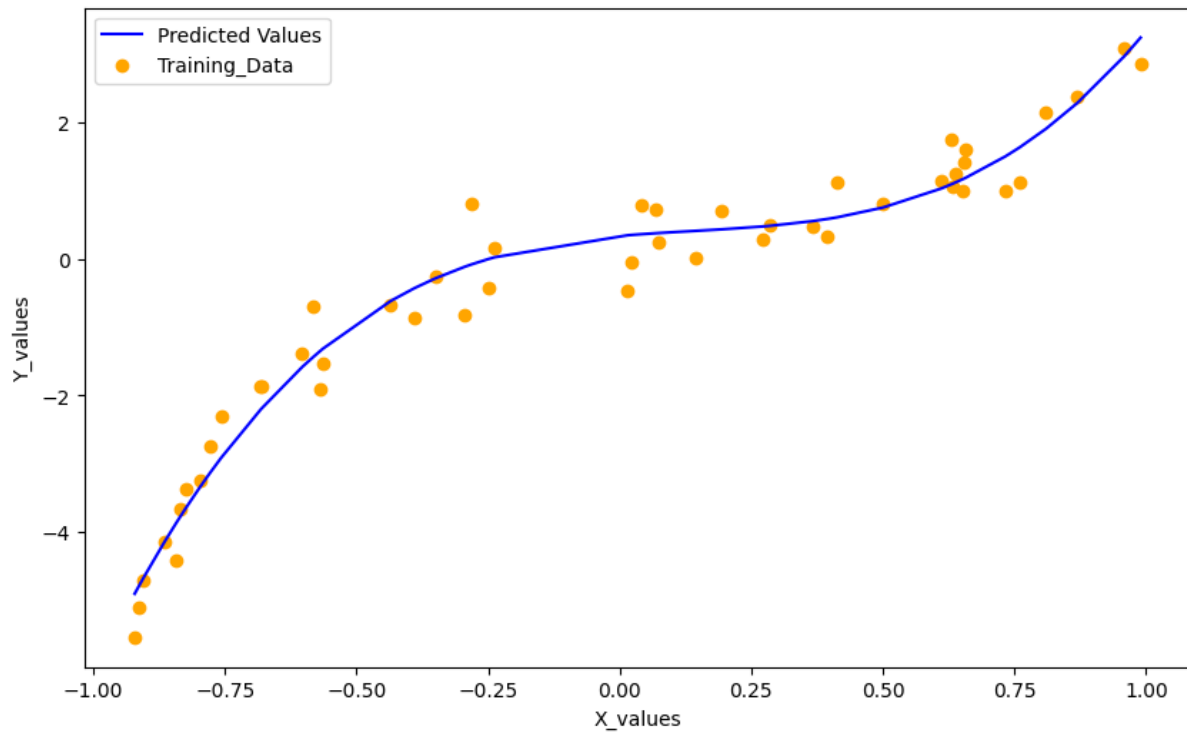


Erms Test: 0.8262755176939132

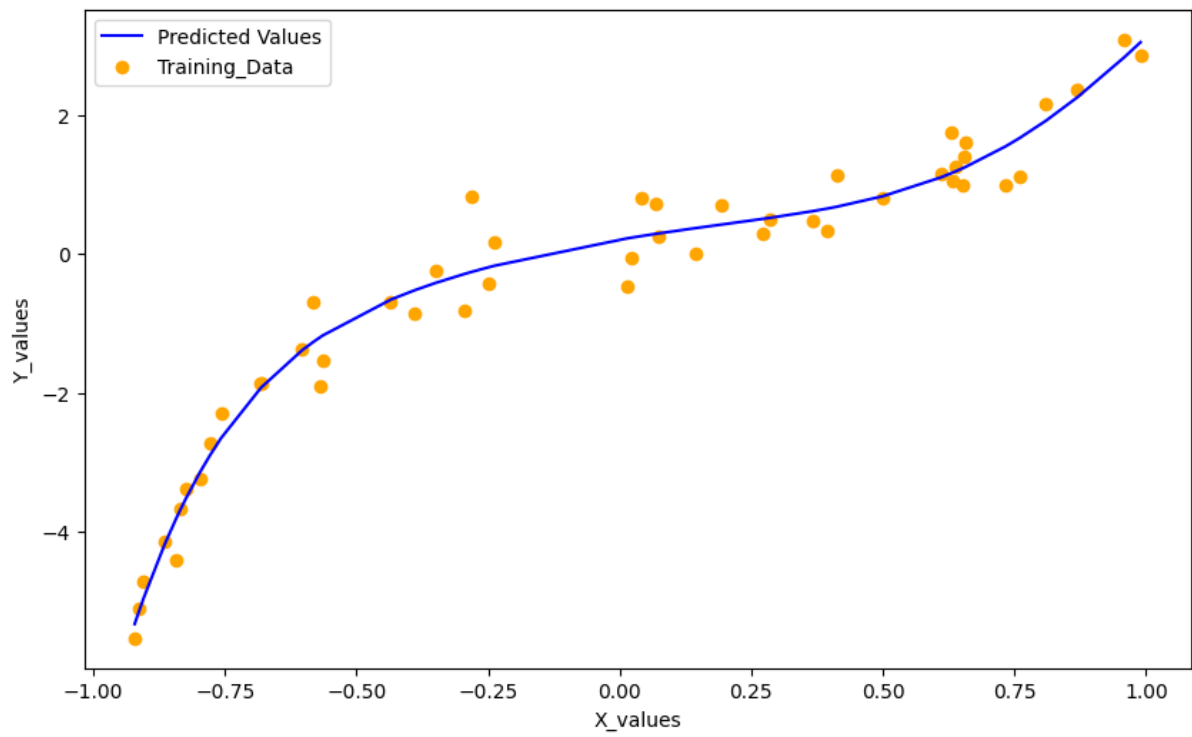
Dataset 1 b:

1. Without Regularization for all degrees :

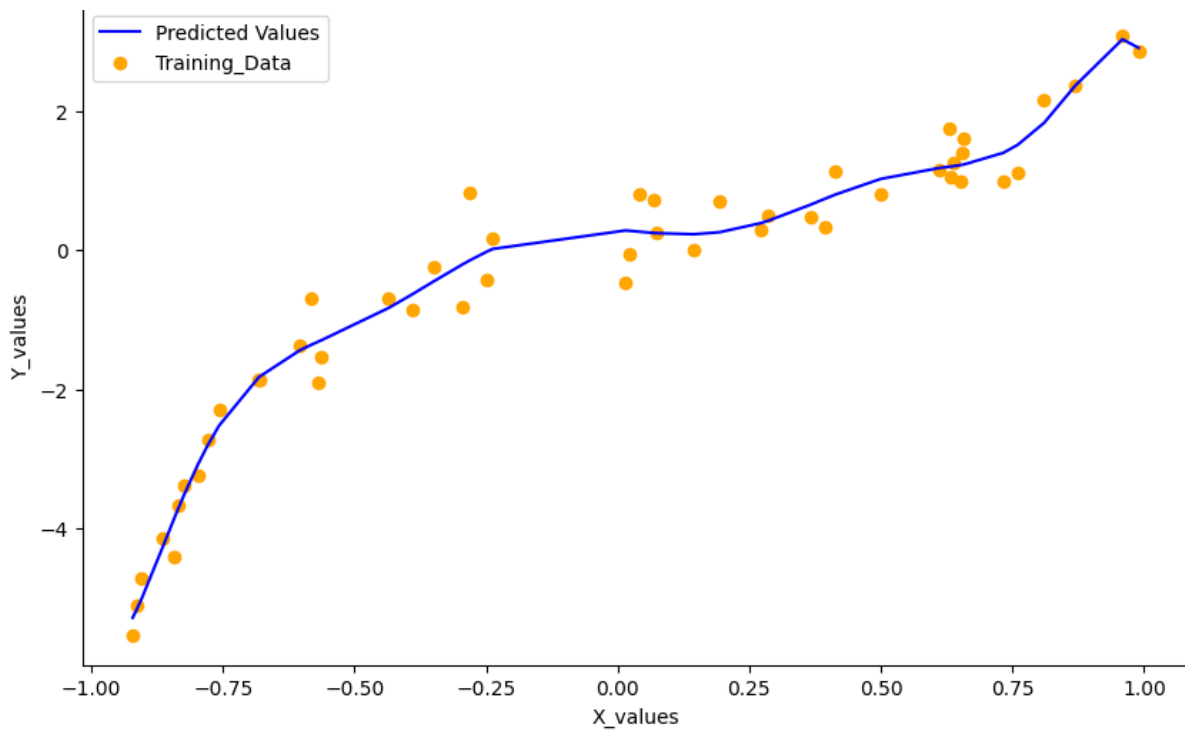
Task 1b size= 50; degree= :3 lamda= : 0



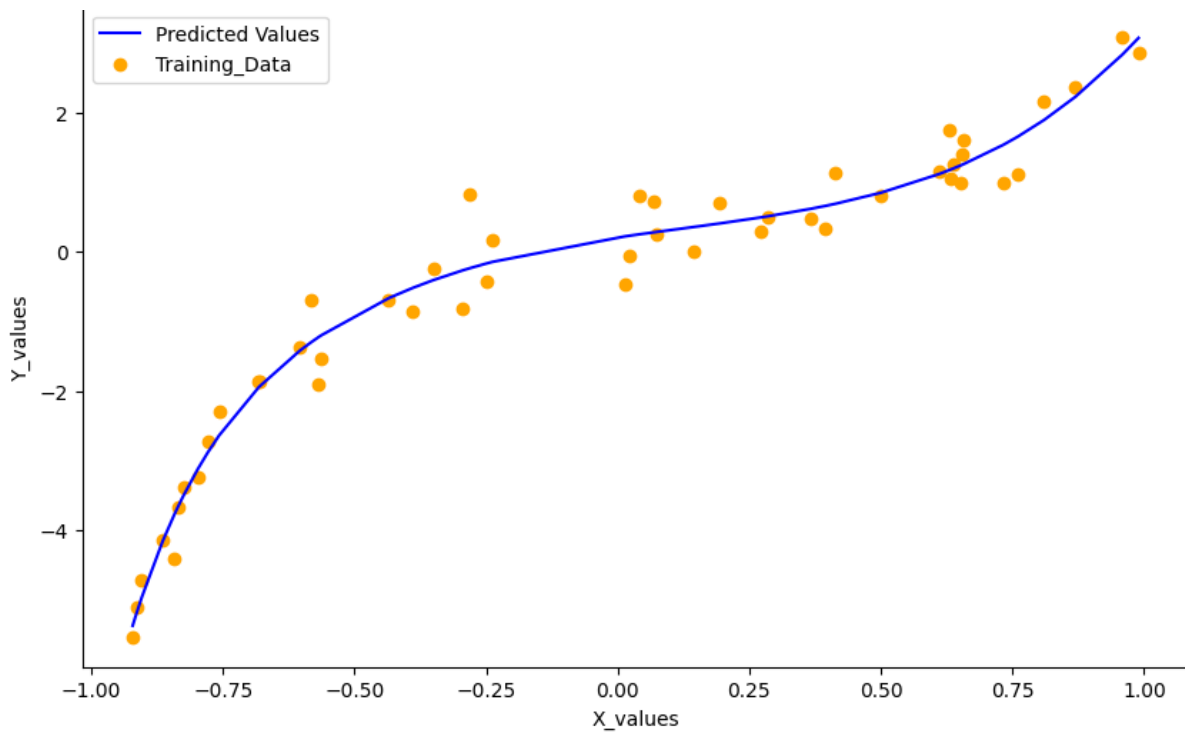
Task-1b size= 50; degree= 6; lamda=0



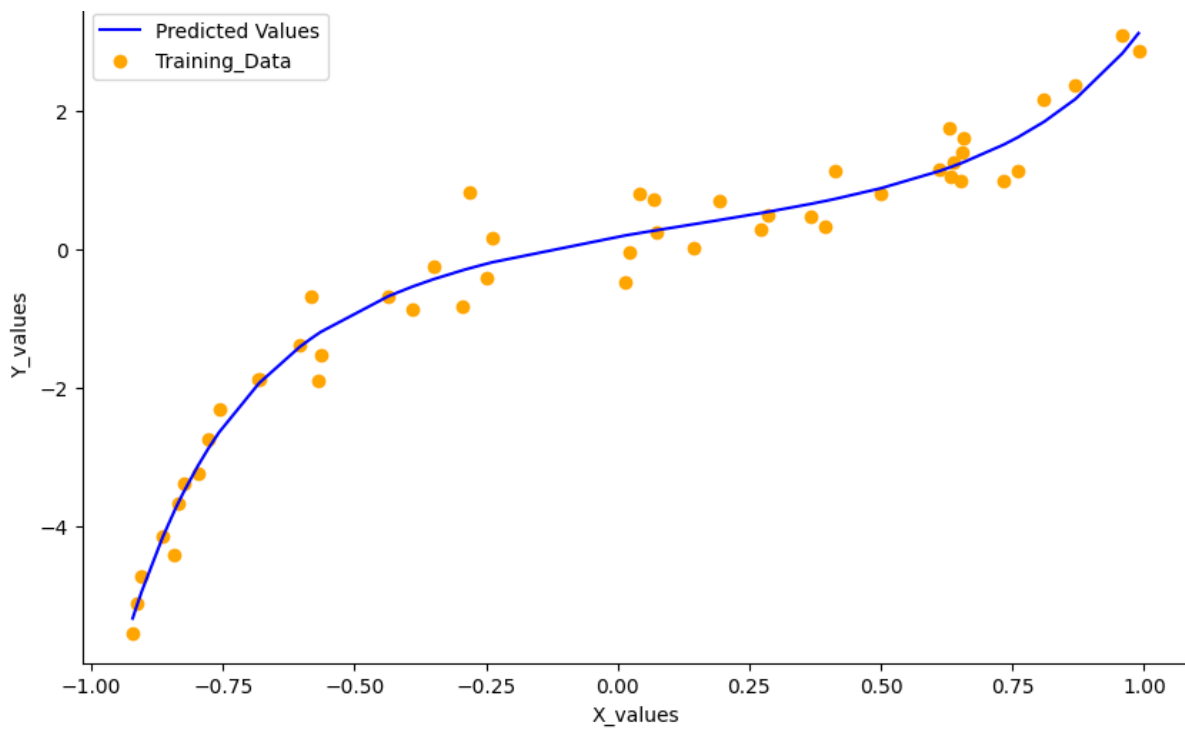
Task 1b size=50; Degree: 9, lamda: 0



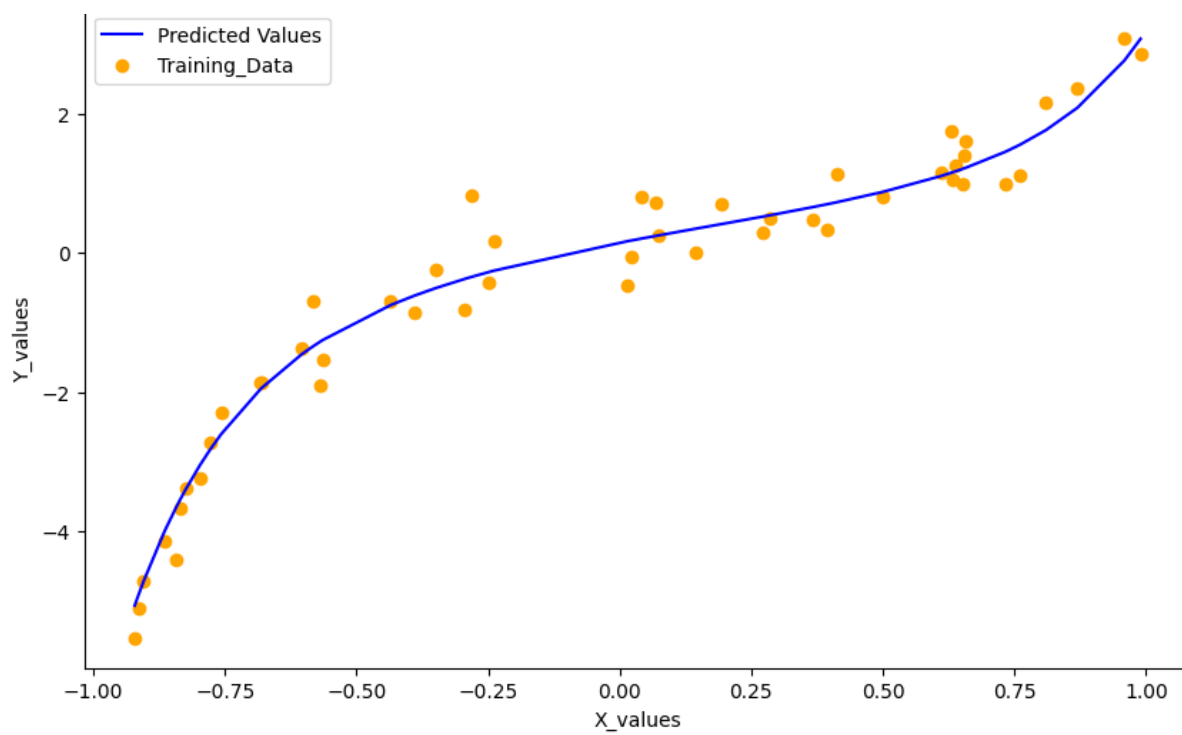
Task-1b size=50; Degree: 9; lamda: 0.001



Task 1b- size=50; Degree: 9, lamda=0.1



Task-1b size=50; degree=9; lamda=1



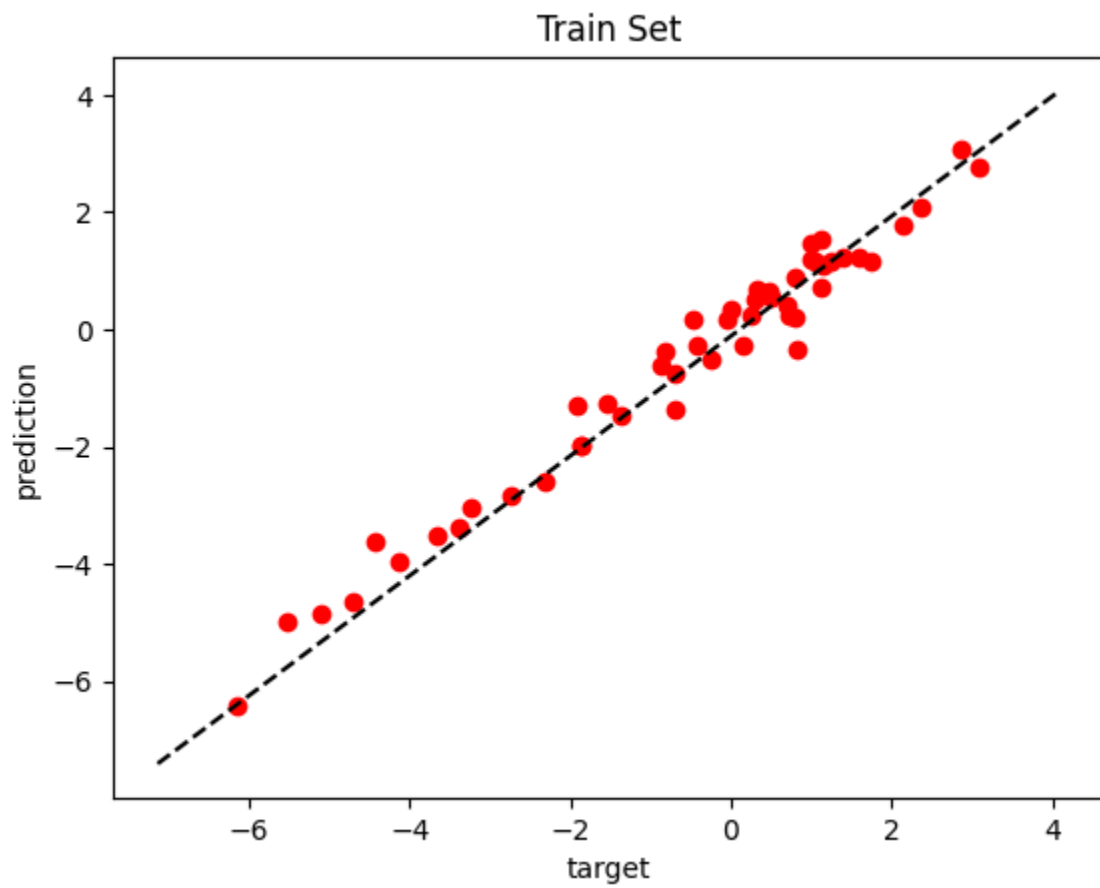
ERMS Table 1B:

	Without Regulization	Without Regulization	Without Regulization	With Regulization with degree=9	With Regulization with degree=9	With Regulization with degree=9
	M=3	M=6	M=9	Lamda=0.001	Lamda=0.1	Lamda=1
Training	0.3923864088658 8727	0.3603079052902 8234	0.3455329606925 022	0.3587877664083 3476	0.3603577095818 529	0.3725666926474 762
Validati on	0.4460346870599 192	0.4460346870599 192	0.4257339786938 165	0.41650078350479 025	0.4125367838889 7946	0.3969372037123 87
Test	0.4525943078073 553	1.1721371936331 701	4.9007721183031 37	0.97845828355580 71	0.4805490714849 7855	0.7465130056268 595

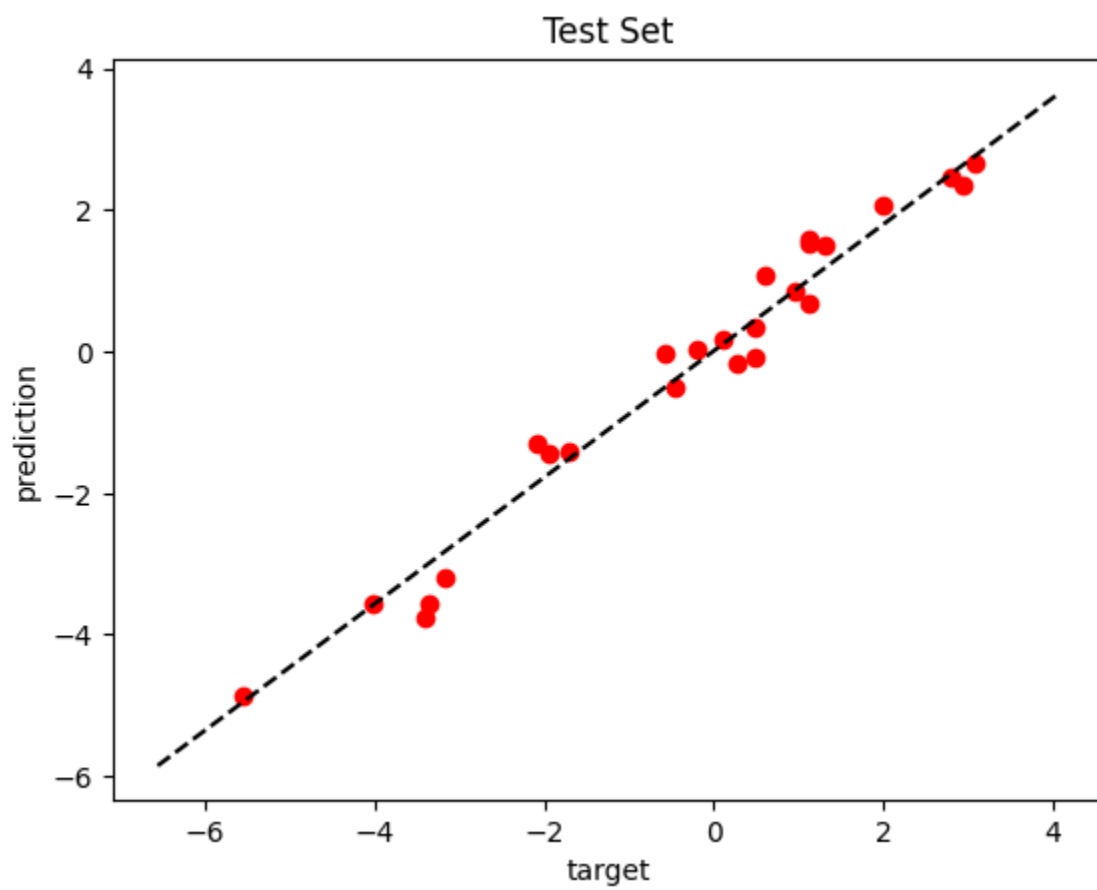
Using above Table best Model with Degree: 9

Lamda: 1 for 1B

Scatter plot for best model



Erms Train: 0.3771



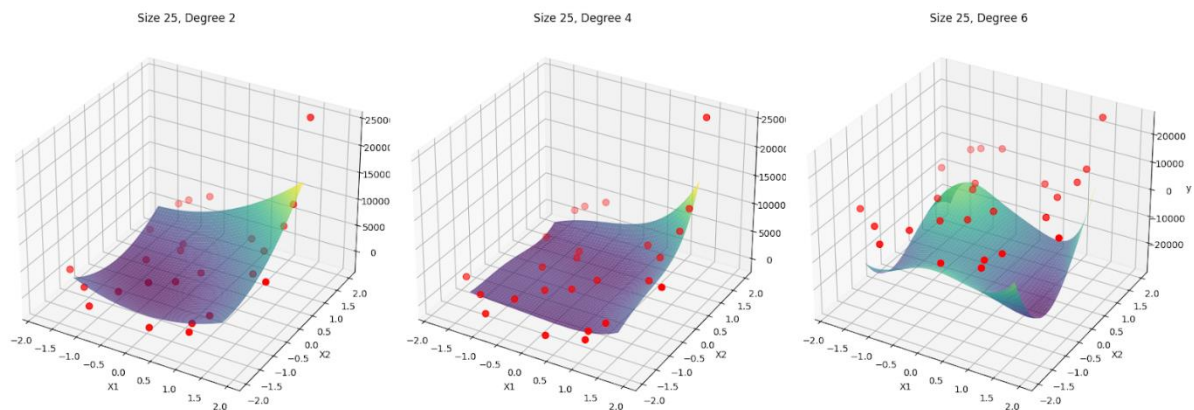
Erms Test: 0.4130

5.2 Dataset 2: Bivariate Data

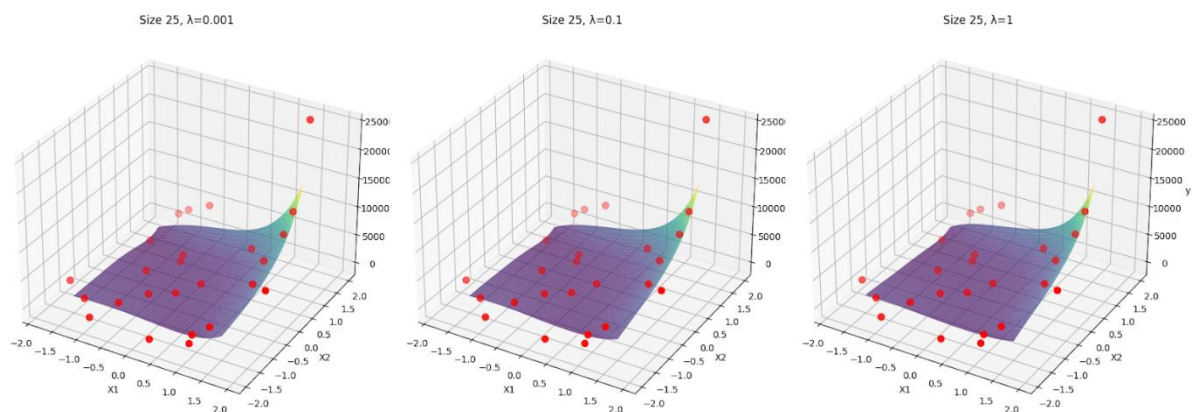
- **Objective:** To plot surfaces of the approximated function obtained using training datasets of different sizes (25 and 100) for different model complexities with and without regularization, and for different values of λ .
- **Plots:** Training data points are superposed on the surfaces.

Dataset 2a: size: 25

Without regularization:



With regularization :



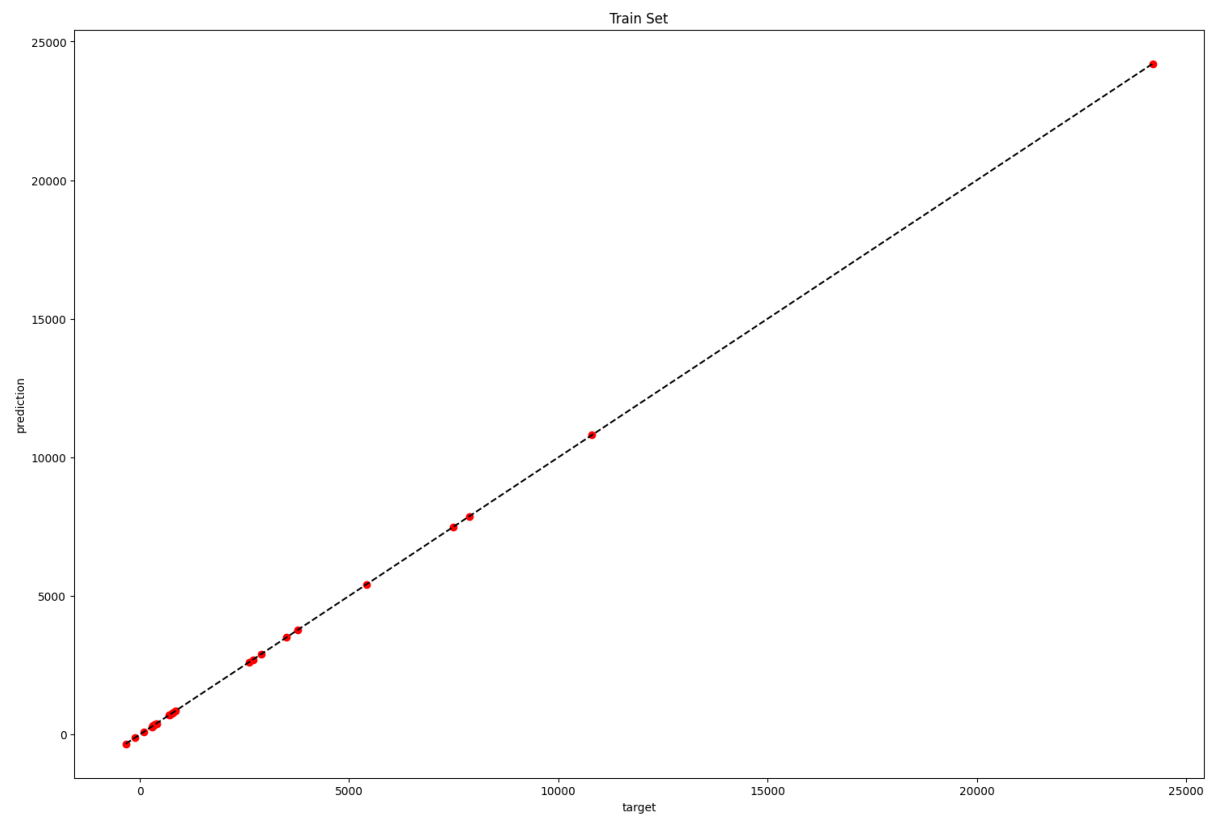
ERMS Table 2A:

	Without Regulization	Without Regulization	Without Regulization	With Regulization with degree=6	With Regulization with degree=6	With Regulization with degree=6
	M=2	M=4	M=6	Lamda=0.001	Lamda=0.1	Lamda=1
Training	2060.6716	172.6635	16674.2394	0.5738	9.6296	43.1166
Validation	1482.1892	306.9111	23562.7387	83.2928	218.6934	361.0791
Test	1900.0157	471.8189	24630.5991	173.4937	480.1068	870.5353

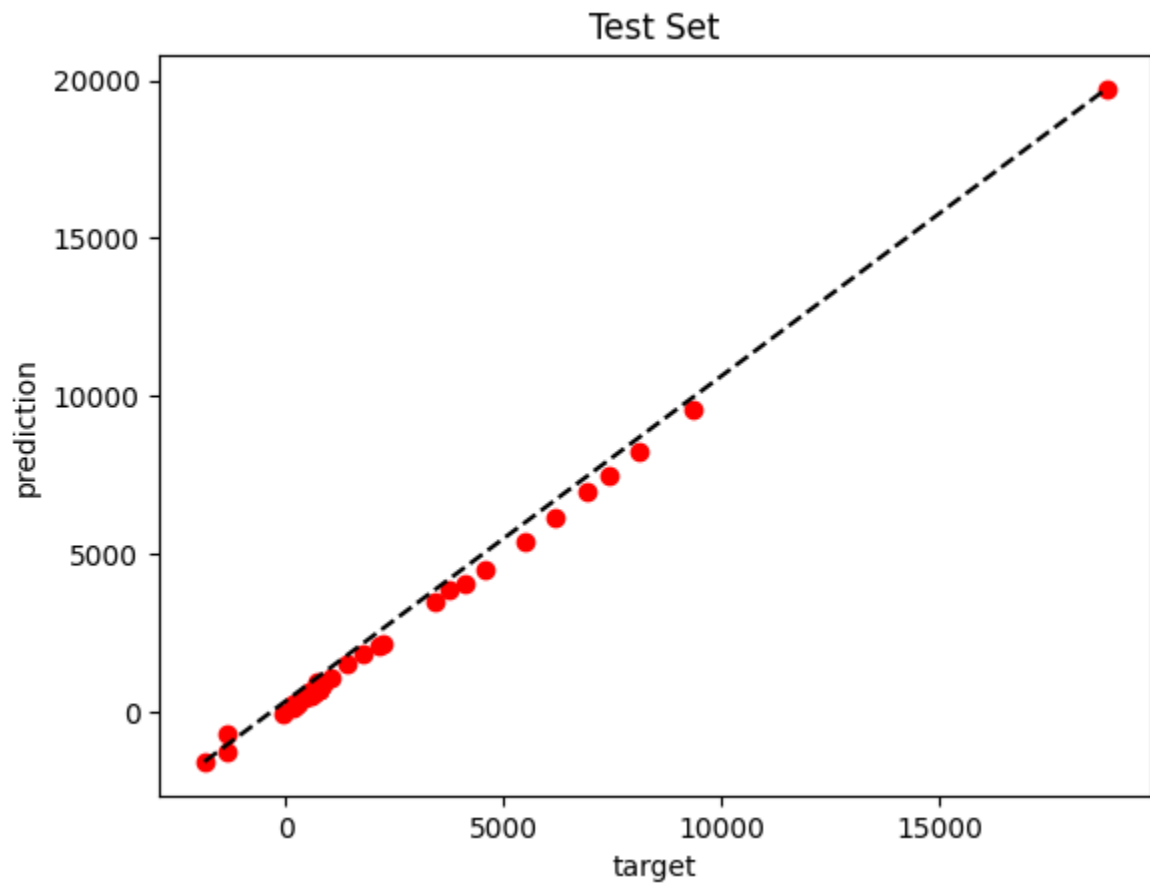
Using above Table best Model with Degree: 4

Lamda: 0.001 for 2A

Scatter plot for best model



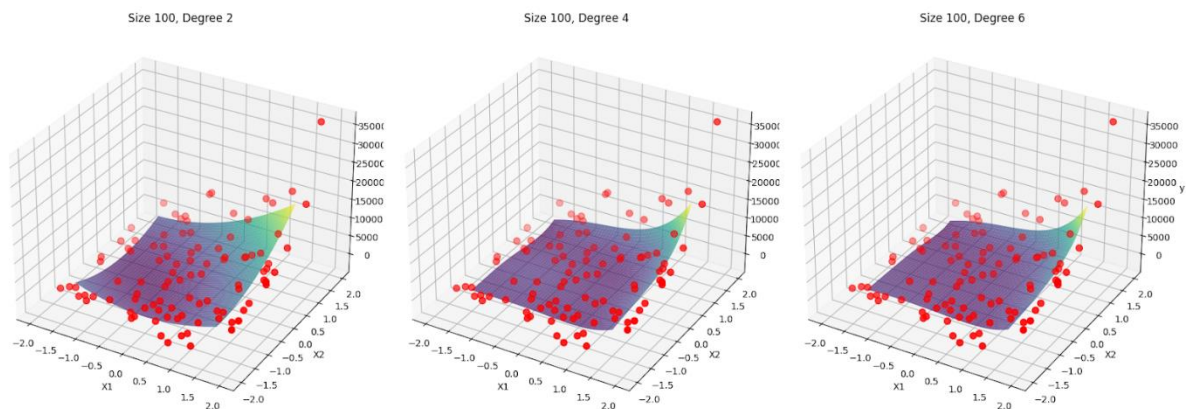
ERms Train: 296.9157



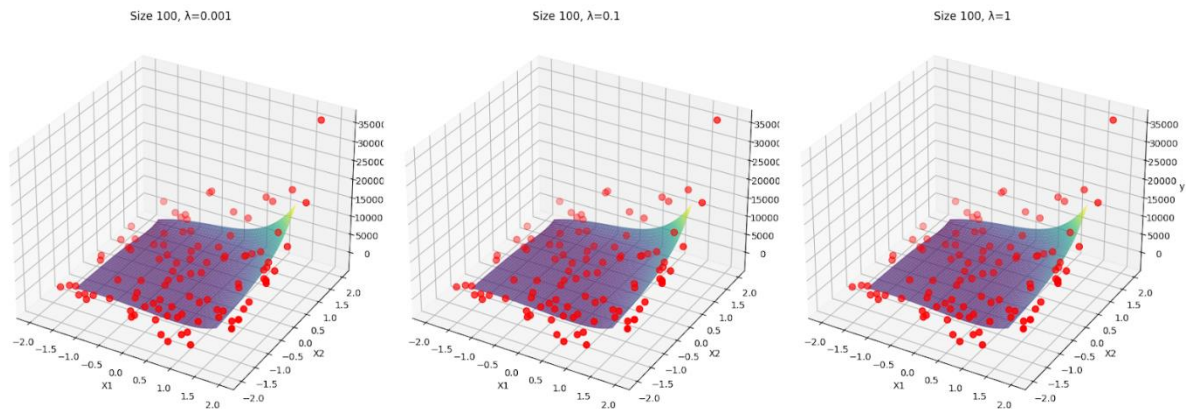
Erms Test: 59.1465

Dataset 2B:

Without regularization:



With regularization:



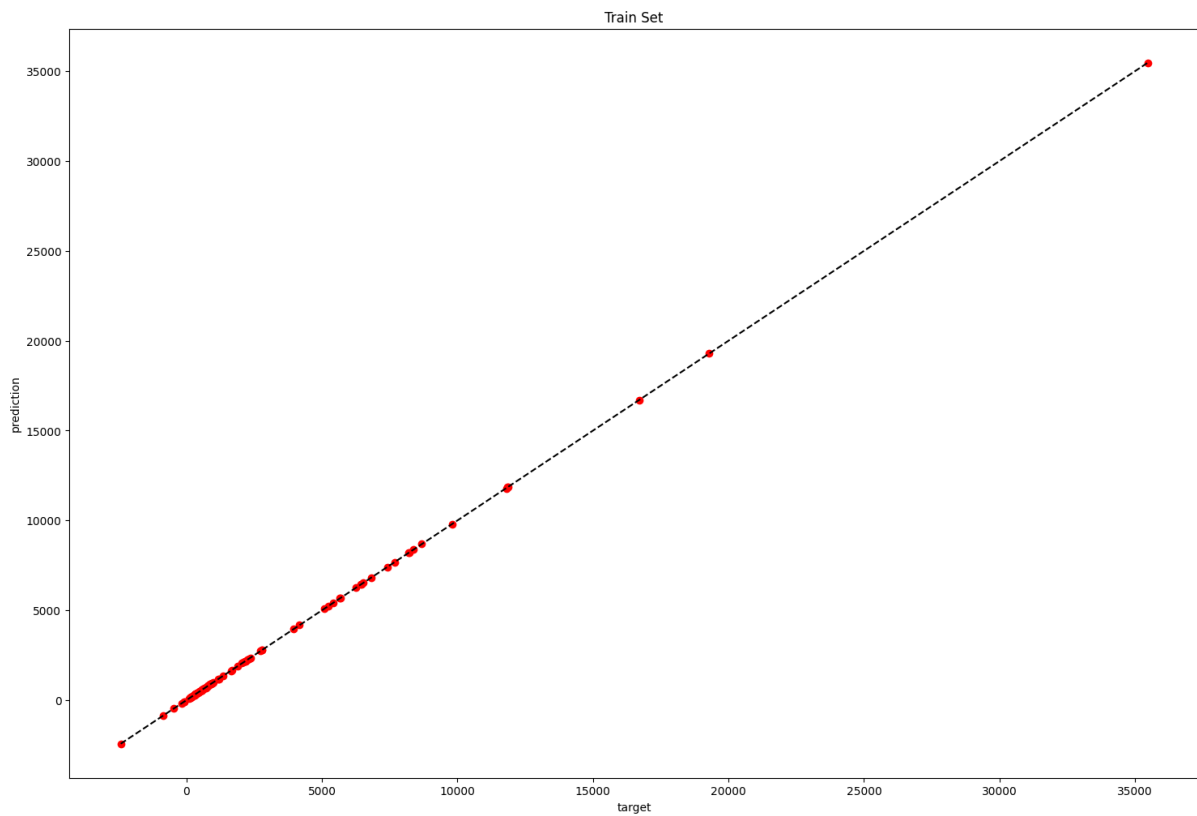
ERMS Table 2B:

	Without Regularization	Without Regularization	Without Regularization	With Regularization with degree=6	With Regularization with degree=6	With Regularization with degree=6
	M=2	M=4	M=6	Lamda=0.001	Lamda=0.1	Lamda=1
Training	2124.7823	367.7128	8.9055	8.9056	9.2486	19.4815
Validation	1612.4302	336.4211	5.5645	5.5566	5.6431	19.8476
Test	1872.4711	420.8524	5.4066	5.3863	4.5209	20.5553

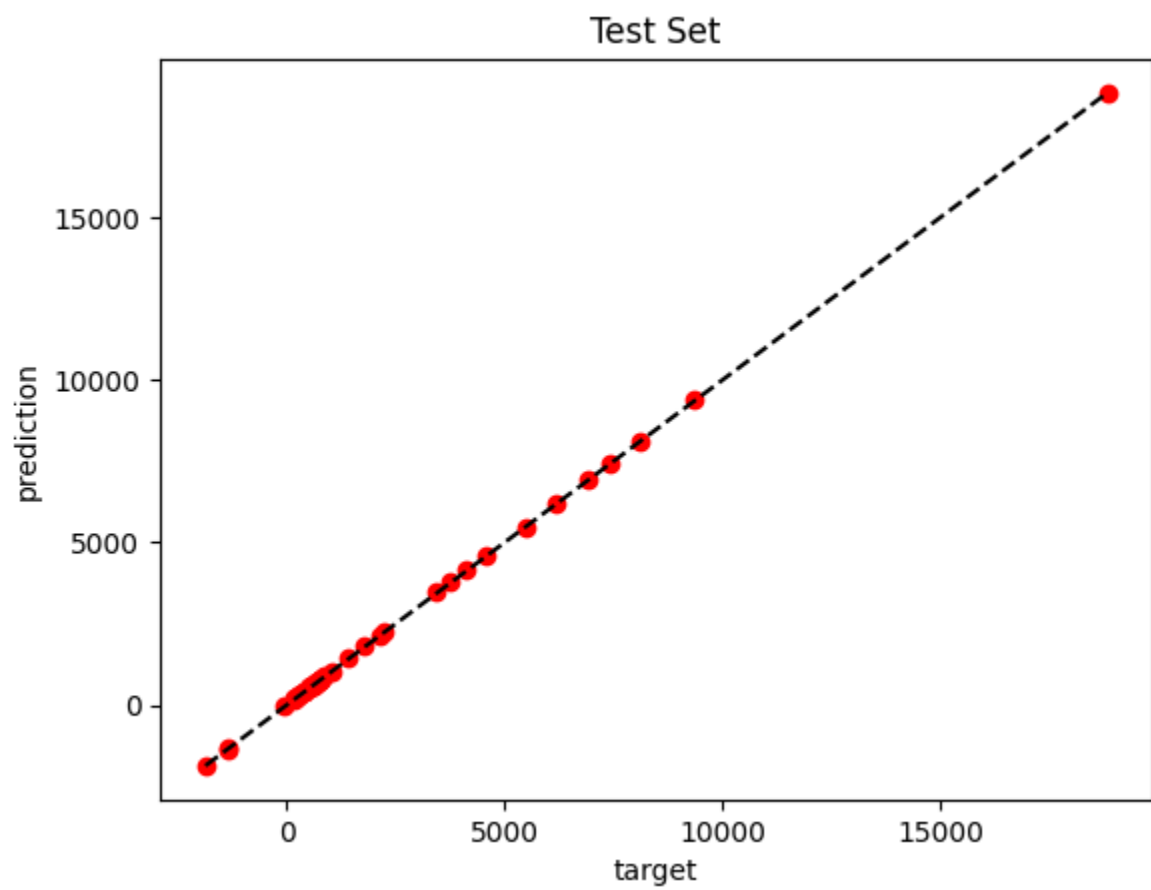
Using above Table best Model with Degree: 6

Lamda: 0.001 for dataset 2B

Scatter plot for best model



Erms train: 8.9056



Erms Test: 5.5566

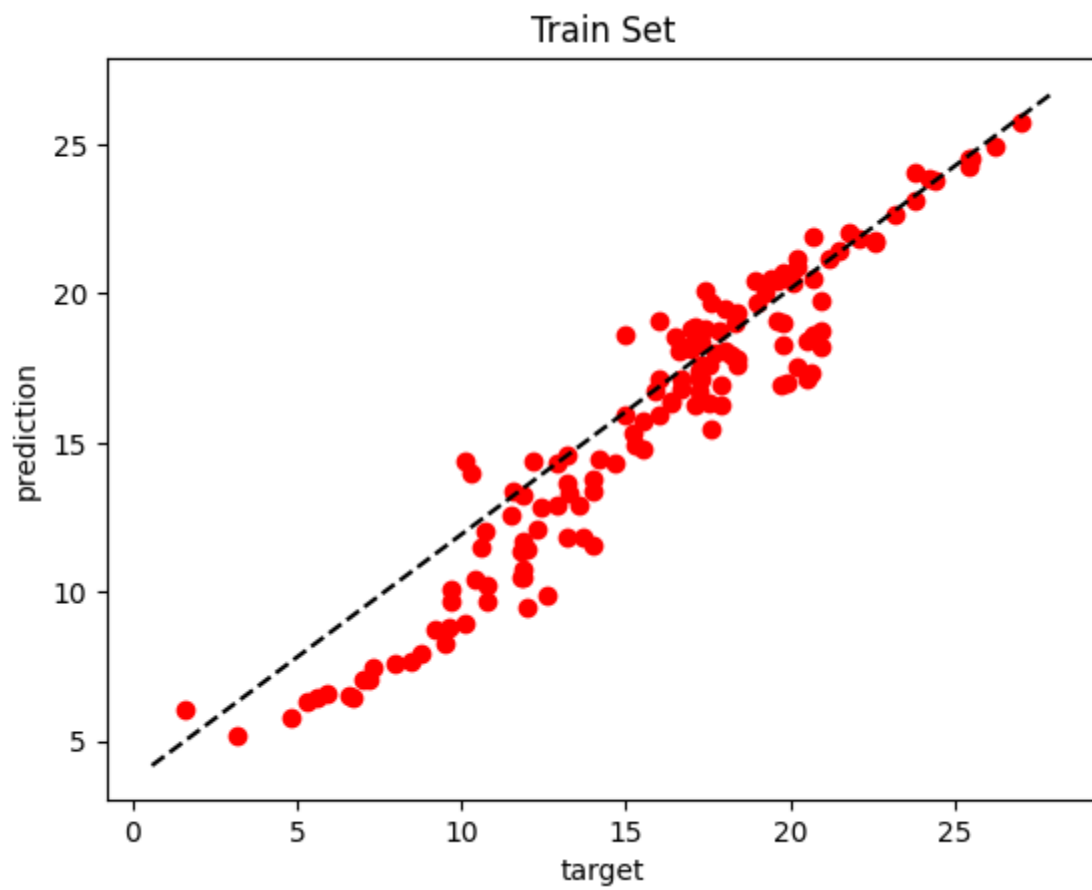
5.3 Data set 3:

	Without Regulization	Without Regulization	With Regulization with degree=3	With Regulization with degree=3	With Regulization with degree=3
	M=2	M=3	Lamda=0.000001	Lamda=0.0001	Lamda=0.1
Training	1.4189	1.303	1.4189	1.303	1.3033
Validation	1.0479	1.8171	1.0479	1.8171	1.8183
Test	1.3206	1.6254	1.3206	1.6254	1.6252

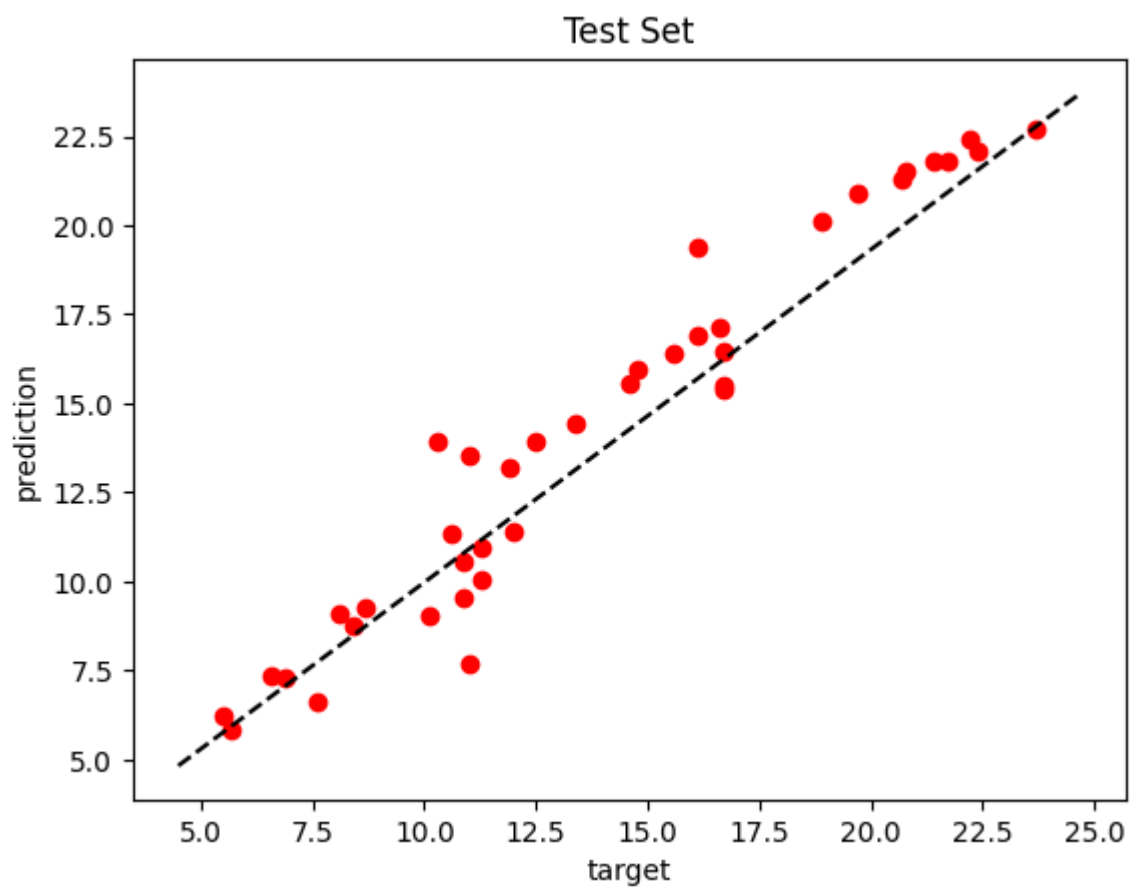
Using above Table best Model with Degree: 2

Lamda: 0.000001 for Dataset 3

Scatter plot for best model



ERms Train:1.3979



Erms Test: 1.2956

Conclusion

Best parameter for ALL dataset

Dataset	Degree	lamda
1A	3	1
1B	9	1
2A	4	0.001
2B	6	0.001
3	2	0.000001

In this assignment, the relationship between model complexity, regularization, and dataset size was explored. The results highlighted the importance of selecting an appropriate degree of polynomial and regularization coefficient to balance the trade-off between bias and variance. The validation method proved effective in identifying the best model, ensuring good generalization performance.